COMPUTER NETWORKS · TRANSPORT LAYER

TCP AND UDP IN COMPUTER NETWORKS

How applications exchange data, and why reliability and latency lead to different choices.

Seminar overview · connection, delivery, and trade-offs

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The transport layer connects applications

IP moves packets between hosts; the transport layer moves data between applications on those hosts. Port numbers identify the sending and receiving applications. TCP and UDP sit above IP and below application protocols.



Application

Web, video, games, DNS



Transport

TCP or **UDP**
Ports and delivery behavior



Network

IP addresses and routing

Application data becomes transport segments or datagrams, then IP packets.

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TCP · TRANSMISSION CONTROL PROTOCOL

RELIABLE DELIVERY

OVERHEAD IS THE TRADE

TCP trades overhead for reliability

Connection-oriented

TCP establishes a connection before application data flows.

Reliable and ordered

Sequence numbers, acknowledgements, and retransmission help deliver complete data in order.

Familiar example

Loading a web page or transferring a file favors correctness over the smallest possible delay.

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02 · TCP CONNECTION SETUP

TCP starts with a three-way handshake

Before data transfer, both endpoints agree to begin a connection and synchronize sequence numbers.

01

CLIENT → SERVER

SYN

Client requests a connection.

02

SERVER → CLIENT

SYN + ACK

Server accepts and acknowledges.

03

CLIENT → SERVER

ACK

Client confirms. Connection Established.

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03 · TCP SEGMENT FORMAT

TCP headers carry delivery control

32-BIT ROW	FIELD	BITS
1	Source Port · Destination Port	16 + 16
2	Sequence Number	32
3	Acknowledgment Number	32
4	Data Offset · Reserved · Control Flags · Window Size	4 + 6 + 6 + 16
5	Checksum · Urgent Pointer	16 + 16
—	Options and Padding, if present	Variable

FIELD PURPOSE

- Sending application · Receiving application
- Order and loss detection
- Next byte expected
- Header size and state · Flow control
- Integrity and urgent data
- Extends and aligns header

MINIMUM HEADER

20 bytes before the Data/Payload.

AFTER THE HEADER

Data / Payload carries application content.

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TCP moves data through the network

ENDPOINT A

CLIENT

Application sends bytes.

DATA →

NETWORK HOP

ROUTER

Forwards IP packets.

→ DATA

ENDPOINT B

SERVER

Receives bytes.

← ACK

The acknowledgement confirms receipt. If data is missing, TCP can retransmit it.

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UDP keeps communication lightweight

UDP is connectionless: independent datagrams need no session.

Low overhead

Small headers and no handshake mean fast setup.

Few built-in guarantees

UDP does not promise delivery, ordering, or retransmission.

Online gaming

Online gaming favors fresh positions over recovered old updates.



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UDP sends datagrams without retransmission

CLIENT

Send

Datagrams leave independently.

DATAGRAM 1 → ROUTER → SERVER

DATAGRAM 2 → ROUTER → SERVER

DATAGRAM 3 → ROUTER ✕ LOST

SERVER

Continue

No built-in retransmission. The application decides what to do next.

The trade-off: less waiting and less overhead, with less delivery control.

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07 · UDP DATAGRAM FORMAT

UDP headers are intentionally small

16 BITS	16 BITS	16 BITS	16 BITS
Source Port	**Destination Port**	**Length**	**Checksum**

8 bytes

MINIMUM HEADER

AFTER THE FOUR FIELDS

Data / Payload

The application data follows immediately. No sequence number, acknowledgement number, or window field is built into the header.

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08 · SIDE-BY-SIDE

TCP and UDP make different trade-offs

CHARACTERISTIC	TCP	UDP
Connection type	Connection-oriented	Connectionless
Reliability and ordering	Built in	Not guaranteed
Acknowledgements	Yes	No built-in ACK
Retransmission	Yes	No built-in retransmission
Minimum header	20 bytes	8 bytes
Overhead	Higher	Lower
Common applications	Web, file transfer	Gaming, live media, DNS

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09 · DESIGN CHOICE

Choose reliability or low latency

Start with what the application can tolerate: missing data, waiting, or both.

01 Does every byte need to arrive?

Yes → TCP
Reliable file transfer and web communication.

02 Is a late update worse than a lost one?

Yes → UDP
Online gaming and live media.

03 Is the exchange short and

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TAKEAWAY

TCP RELIABILITY

ORDER · ACKNOWLEDGEMENT · RECOVERY

TCP is reliability-focused; UDP is latency-friendly

TCP pays for dependable, ordered delivery. UDP keeps the path lightweight and fast, leaving more decisions to the application.

THANK YOU

ANY QUESTIONS?

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